

Your updated full review pack

14 evidence cards | Performance opportunity-v3 | 25 September 2026

One decision per card. Everything needed for the selected checks is printed here; opening WSPRadar is optional. Longer cards continue onto the next page so that the source rows, station roles and explanations stay readable.

Time budget: approximately 41 minutes for the cards, plus the revision overview. Stop after one hour; mark any unresolved card Unclear / skip.

Print on A4 at actual size. A pen is enough; a basic calculator is optional. Check the selected anchors; you do not need to recompute every decimal or inspect every archive row.

- 1. Review.** Read the role box and evidence table or marked figure, follow the numbered checks, and compare the proposed answer with the WSPRadar result.
- 2. Mark.** Tick Verified only if the evidence and arithmetic support that anchor. Tick Discrepancy and write a correction if needed. Blank or unclear cards remain unapproved.
- 3. Photograph.** Include the complete card, including any continuation page and its decision boxes. Send cards in any order. No transcription or separate app screenshots are required.

WHAT YOUR TICK APPROVES

The specific printed anchor and its interpretation, not every value in the run. Approved anchors can then be frozen; complete-row and consistency checks provide wider automated regression coverage.

REVIEW HISTORY PRESERVED

B01's prior numeric approval is retained. P01 was approved under opportunity-v2 and needs renewed review under the revised method. All other cards remain unapproved. This packet records no new human decisions.

REVIEW ORDER

Card	Check	Minutes
B01	One pair: normalization and Delta SNR	2
P01	Hit, Miss, Target-only - and excluded cycles	3
P02	VE6PDQ: count the right opportunities	3
P03	Successful SNR: median and extremes	3
P04	Map weighting: five unequal stations	3
P05	One distance bin, three related figures	3
P06	Peer Reach includes zero-Hit stations	3
P07	One complete three-hour bin	3
P08	UTC folding at the end of the window	3
P09	Folded SNR: each date-hour gets one vote	3
P10	SNR deviation: subtract each station baseline	3
B02	Histogram weights and a positive outlier	3
B03	Decode Outcomes: which denominator?	3
B04	Map value: two stations, three observations	3

Revised candidate evidence | Existing human decisions preserved

What changed - and why

COMMON SETUP AND STATION ROLES

TX analysis on 40 m. Target TX: KP4MD / CM98. Selected peer RX: VE6PDQ / DO34IR; population cards name the other peer receivers. Benchmark Reference TX: WB6RQN / CM98, no correction. Window: 19 Dec 2010 12:00 to 20 Dec 2010 20:00 UTC (end exclusive). Radius strictly below 5,000 km; all hours and directions; special callsigns and moving stations included. Performance minimum: 5 opportunities. Benchmark minimum: 1 Joint Spot.

REVISED PERFORMANCE CLASSIFICATION

KP4MD (Target TX) -> VE6PDQ (peer RX): a valid Target decode confirms both endpoints participated. An external TX -> VE6PDQ report confirms that VE6PDQ was listening. KP4MD -> another RX confirms only the Target TX's activity for the silent VE6PDQ path; it does not establish that VE6PDQ was listening.

Retained same-band, same-cycle evidence	Classification
Target decoded, with or without external confirmation	Hit / success; one opportunity
External confirmation; no Target decode	Miss; one opportunity
Insufficient endpoint activity evidence	Unknown; excluded
Target decoded without external confirmation	Target-only subset of successes; never add it again

The Target-Active Gate, exact peer callsign/full-locator identity, strongest-report consolidation, reported-power normalization, filters, Benchmark and Delta-SNR methods remain unchanged.

ACTUAL BEFORE / AFTER RESULTS

Quantity	opportunity-v2	opportunity-v3
VE6PDQ successes / opportunities	35 / 57	57 / 79
VE6PDQ Decode Rate	61.4035%	72.1519%
VE6PDQ Misses	22	22
VE6PDQ successful-SNR median	-25 dB	-23 dB
Receivers passing minimum 5	82	85
Population station-balanced rate	37.1797%	42.5054%
Population pooled rate	33.8311%	41.7555%
Qualified successes / opportunities	725 / 2143	1018 / 2438

Newly qualifying peer RX: K7LRB / DM41MW, NN6RF / CM87UW and VE7VI / CN89FJ. Qualified-population Misses grow from 1,418 to 1,420 because VE7VI brings two existing Misses into that population. Across all 102 peer identities, 1,445 Misses remain unchanged.

P01-P07 and P09-P10 have numeric changes. P08's selected 12/20 UTC anchors remain unchanged. B01-B04 retain their Benchmark calculations. Every card now states the station roles and transmission direction.

EVIDENCE AND VERIFICATION

Fresh generated strict/legacy SQL and production processing were replayed against the frozen 65,647-report source. All 2,478 native peer-cycle rows, station thresholds, map segments and prepared distance/temporal profiles agree with an independent raw-source oracle. The compact figures on these cards are newly rendered from those production recipes. Benchmark publication fixtures and their numeric card anchors remain unchanged.

The replay uses the regression SQLite dialect adapter, not a native ClickHouse service. This historical TX packet does not independently validate RX; separate hand-checked RX/TX contracts cover that method and duplicate reduction. The implementation suite passed: 3,179 passed, 1 skipped, 3 expected failures. Automated agreement is not human approval.

Frozen source SHA-256: 602e3acc6db36e6fc8e1ba00760218dbef2a1bb94679b0565295f8edaea7aa6e
Configuration SHA-256: 0ed68b9a16e7b3a944d7c183a31033d25eca636b7321f1c952b0cb56ab2d39c6

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

One pair: normalization and Delta SNR

Prior numerical approval retained from B01; this replacement has no new human decision.

Analysis Direction: TX Benchmark

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6PDQ / DO34IR

Reference Tx: WB6RQN / configured locator CM98 (raw reports CM98MQ)

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

Reference TX WB6RQN -> peer RX VE6PDQ / DO34IR

Benchmark | Common setup on cover. VE6PDQ (DO34ir), 20 Dec 00:22 UTC.

Activity evidence: Each Target or Reference decode confirms its named TX transmitted and that exact peer RX received. Joint pairs require both TX reports at the same peer RX/full locator/cycle; unchanged Benchmark rules apply.

Revision change: Benchmark numerical anchors and methods unchanged; endpoint roles added.

Report ID	TX callsign / locator	RX callsign / locator	Raw SNR dB	Power dBm	At 30 dBm
44698904	KP4MD / CM98IQ	VE6PDQ / DO34IR	-13	37	-20
44698905	WB6RQN / CM98MQ	VE6PDQ / DO34IR	-15	33	-18

WHAT TO CHECK

1. Read the two raw report rows. Subtract the reported power in dBm, then add 30 dBm, for each SNR.
2. Subtract Reference from Target. Tick only if the -2 dB result and direction are clear.

Proposed answer: KP4MD: -13 - 37 + 30 = -20 dB. WB6RQN: -15 - 33 + 30 = -18 dB.

Delta SNR = -20 - (-18) = -2 dB: Reference is 2 dB stronger after normalization.

The same receiver, full locator and cycle create one Joint observation, not two independent differences.

WSPRadar result: Target -20 dB; Reference -18 dB; one Joint observation at -2 dB.

Protects: Detects sign reversal, omission of reported-power normalization and incorrect evidence units.

Notes: Power values are the archived dBm fields: 37 dBm is about 5 W and 33 dBm about 2 W. This card checks dBm arithmetic, not rounded watt conversions.

Source: Frozen Milazzo raw reports; independent arithmetic and production replay. Full ledgers are retained with this review pack.

YOUR DECISION - B01 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Hit, Miss, Target-only - and excluded cycles

Original-method example approved; revised classification requires fresh human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | VE6PDQ / DO34IR; three retained examples and two excluded cycles. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: Old examples: 1 Hit / 2 opportunities = 50%. Revised: 2 Hits / 3 opportunities = 66.6667%; 1 Miss unchanged.

Report ID	UTC	TX callsign / locator	RX callsign / locator	Activity directly confirmed
44619841	19 Dec 14:06	KP4MD / CM98IQ	VE6PDQ / DO34IR	Target TX KP4MD + peer RX VE6PDQ
44619843	19 Dec 14:06	N4AU / EM62VP	VE6PDQ / DO34IR	Supporting TX N4AU + peer RX VE6PDQ
44619818	19 Dec 14:06	KP4MD / CM98IQ	NW0Q / EM38TV	Target TX KP4MD + other RX NW0Q
44646805	19 Dec 18:00	N6RZR / CN80VK	VE6PDQ / DO34IR	Supporting TX N6RZR + peer RX VE6PDQ
44646808	19 Dec 18:00	7L4IOU / PM95WR	VE6PDQ / DO34IR	Supporting TX 7L4IOU + peer RX VE6PDQ
44646786	19 Dec 18:00	KP4MD / CM98IQ	WL7BDO / BP54KN	Target TX KP4MD + other RX WL7BDO
44618722	19 Dec 13:58	KP4MD / CM98IQ	VE6PDQ / DO34IR	Target TX KP4MD + peer RX VE6PDQ
44605242	19 Dec 12:06	WD6DBM / CM97AI	VE6PDQ / DO34IR	Supporting TX WD6DBM + peer RX VE6PDQ; no Target TX witness
44629351	19 Dec 15:20	KP4MD / CM98IQ	K6HX / CM87	Target TX KP4MD + other RX K6HX; VE6PDQ activity unknown

19 Dec UTC	Target seen	External seen	Outcome / opportunity
14:06	yes	yes	Hit / yes
18:00	no	yes	Miss / yes
13:58	yes	no	Hit (Target-only subset) / yes
12:06	no	yes	Excluded: Target-Active Gate fails
15:20	no	no	Excluded: peer RX activity unknown

WHAT TO CHECK

1. At 14:06, KP4MD -> VE6PDQ directly confirms Target TX and peer RX participation. The N4AU -> VE6PDQ report adds external provenance; a Hit does not require it.

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

2. At 18:00, N6RZR and 7L4IOU -> VE6PDQ confirm this peer RX received, while KP4MD -> WL7BDO confirms Target TX activity. No KP4MD decode at VE6PDQ gives one Miss.

3. At 13:58, KP4MD -> VE6PDQ alone confirms both endpoints: one Target-only Hit. At 12:06 no KP4MD report exists anywhere that cycle, so the Target-Active Gate excludes the WD6DBM report.

4. At 15:20, KP4MD -> K6HX confirms Target TX activity, but no report was received by VE6PDQ/DO34IR that cycle. This silent peer RX remains unknown and excluded, not a Miss.

Proposed answer: Three retained examples: 2 Hits, 1 Miss, 3 opportunities; $100 \times 2/3 = 66.6667\%$. One Hit has Target-only provenance; do not add it again.

This three-row rate illustrates the arithmetic; it is not the full-window VE6PDQ rate.

WSPRadar result: Production classifies 14:06 H, 18:00 M and 13:58 H (Target-only subset). The 12:06 cycle without Target TX activity evidence and the unknown-VE6PDQ-activity path at 15:20 are excluded.

Protects: Separates Target TX activity from exact peer RX activity, and counts a valid direct decode as success without an unnecessary external-witness requirement.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P01 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

VE6PDQ: count the right opportunities

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | Full 32-hour window; VE6PDQ / DO34IR. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 35/57 = 61.4035% -> 57/79 = 72.1519%; 22 Misses unchanged.

Evidence subset	Count	In totals?	Meaning for peer RX VE6PDQ
Externally supported Hits	35	Yes: Hits + opportunities	Target TX KP4MD and another TX decoded
Target-only Hits	22	Yes: already a Hit subset	Target TX KP4MD decoded; no external report
All Hits	57	57 successes	35 + 22, not 57 + 22
Misses	22	22 opportunities	Other TX decoded here; Target TX active
All opportunities	79	79 = 57 + 22	One count per exact peer RX/cycle

WHAT TO CHECK

1. Add the disjoint success subsets: 35 externally supported + 22 Target-only = 57 Hits.
 2. Add Hits and Misses once: 57 + 22 = 79 opportunities, meeting the minimum of 5.
 3. Calculate 100 x 57/79. Target-only is already inside 57; adding it again would double-count successes.
- Proposed answer:** Expected station rate: 72.1519%, displayed 72.2%.

All 79 retained receiver-cycles enter the denominator; 22 Misses remain unchanged.

WSPRadar result: Station Insights and Drill-Down: 57 Hits, 22 Misses, 79 opportunities, 72.2%; 22 Target-only observations remain a provenance subset of Hits.

Protects: Protects station counts, global eligibility and the most important Performance denominator.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P02 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason:

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Successful SNR: median and extremes

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | VE6PDQ / DO34IR; all 57 successful Target decodes. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 35 -> 57 successful samples; median -25 -> -23 dB; IQR [-28,-22] -> [-29,-22]; extremes [-33,-18] -> [-35,-14] dB.

Rank	Report ID	UTC	TX callsign	RX callsign	Raw SNR	Power dBm	At 30 dBm
1	44644486	19 Dec 17:36	KP4MD	VE6PDQ	-28	37	-35
15	44776098	20 Dec 17:12	KP4MD	VE6PDQ	-22	37	-29
29	44627774	19 Dec 15:06	KP4MD	VE6PDQ	-16	37	-23
43	44773164	20 Dec 16:36	KP4MD	VE6PDQ	-15	37	-22
57	44618722	19 Dec 13:58	KP4MD	VE6PDQ	-7	37	-14

WHAT TO CHECK

1. For each row calculate normalized SNR = reported SNR - reported power + 30. All five reports are Target TX KP4MD/CM98IQ -> peer RX VE6PDQ/DO34IR.
2. With 57 sorted Hits, the median is rank 29. Linear Q1 and Q3 fall exactly at ranks 15 and 43.
3. Include all 57 successful Target decodes, including the 22 Target-only successes. Exclude Misses from successful-SNR summaries.

Proposed answer: Median -23 dB; Q1 -29 dB; Q3 -22 dB; minimum -35 dB, maximum -14 dB.

WSPRadar result: Production successful-SNR summaries match median -23 dB, IQR [-29, -22] dB and range [-35, -14] dB.

Protects: Tests absolute normalized SNR separately from Delta SNR and includes the complete successful-decode population.

Source: Ranks independently rebuilt from raw Target TX reports, sorted by normalized SNR then UTC. Raw locators: KP4MD/CM98IQ -> VE6PDQ/DO34IR. Quartiles are a ledger check; Station Insights displays the median. Temporal IQR is checked in P07/P10.

YOUR DECISION - P03 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason:

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Map weighting: five unequal stations

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: K5XL / EM12KP; KE7A / EM12KX; N5WY / EM12PU; NW0Q / EM38TV; WA5M / EM22JP

Target TX KP4MD -> peer RX K5XL, KE7A, N5WY, NW0Q, WA5M

TX Performance | Map segment [0-2500 km] East; minimum 5 opportunities per station. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 49/97 -> 56/104; station-balanced 63.9686% -> 65.8528%; pooled 50.5155% -> 53.8462%.

Peer RX / locator	Hits	Misses	Opportunities	Rate %
K5XL / EM12KP	4	2	6	66.6667
KE7A / EM12KX	6	4	10	60.0
N5WY / EM12PU	25	30	55	45.4545
NW0Q / EM38TV	16	12	28	57.1429
WA5M / EM22JP	5	0	5	100.0

WHAT TO CHECK

1. Compute the five exact ratios: 4/6, 6/10, 25/55, 16/28, 5/5.
2. The map gives each qualifying peer RX one equal vote: average those five exact ratios.
3. For the pooled diagnostic, add 56 Hits and 48 Misses, then calculate 56/104.

Proposed answer: Map station-balanced value = 65.8528% (displayed 65.9%).

Pooled value = 53.8462% (displayed 53.8%).

WSPRadar result: The East map segment has 5 qualifying peer RX identities, 56 Hits and 104 opportunities: map value 65.9%, pooled diagnostic 53.8%.

Protects: A 12.01 percentage-point difference catches accidental pooling or weighting by observation count.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P04 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

One distance bin, three related figures

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

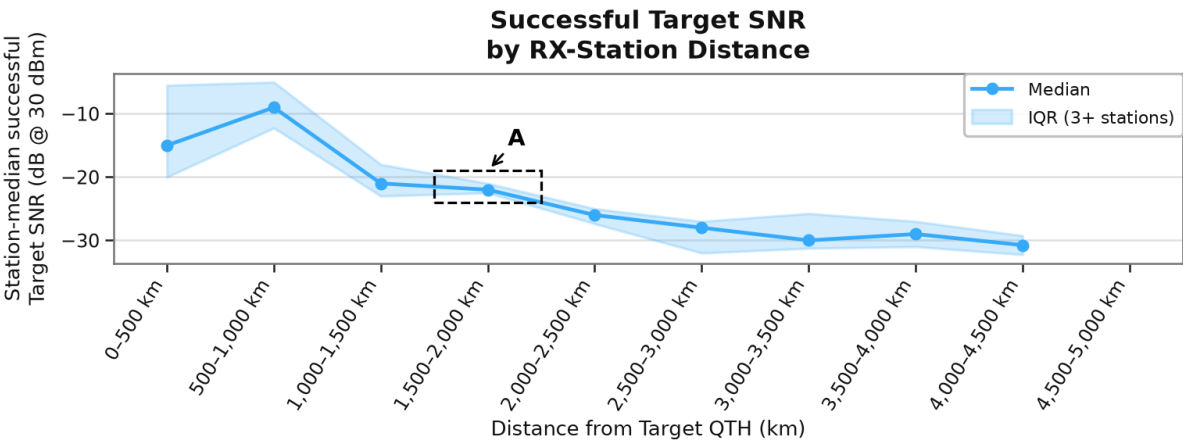
Peer Rx: VE6OG / DO33; VE6PDQ / DO34IR; W6YQ / DN84IC

Target TX KP4MD -> peer RX VE6OG, VE6PDQ, W6YQ

TX Performance | All directions; exact distance 1500-2000 km. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 50/78 -> 75/103; station-balanced 65.3396% -> 71.4580%; pooled 64.1026% -> 72.8155%. Reach and median unchanged; IQR updated.



A marks the review area. Production plot elements; layout adapted for this card.

Peer RX / locator	km	Hits	Misses	Opportunities	Rate %	Successful SNR median dB
VE6OG / DO33	1776.0	13	2	15	86.6667	-20
VE6PDQ / DO34IR	1895.5	57	22	79	72.1519	-23
W6YQ / DN84IC	1601.0	5	4	9	55.5556	-22

WHAT TO CHECK

1. All three peer RX identities qualify and have at least one Hit, so Peer Reach is 3/3 = 100%.
2. Average 13/15, 57/79 and 5/9 for the station-balanced rate; compare with pooled 75/103.
3. Sort station SNR medians -23,-22,-20. Their median is -22; interpolated quartiles are -22.5 and -21.

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Proposed answer: Reach 100%; station-balanced 71.4580%; pooled 72.8155%.

Distance SNR trace -22 dB, IQR [-22.5, -21] dB from three station medians.

WSPRadar result: The three native distance panels match: 100% reach; 71.46% and 72.82% rates; median -22 dB and IQR [-22.5, -21] dB.

Protects: Links all three distance panels to the same independently checked station set and protects station-balanced SNR aggregation.

Source: Performance distances use reported locator centers. Benchmark distances use archived peer coordinates; small cross-mode distance differences do not change membership here.

YOUR DECISION - P05 / revision 2

☐ Verified ☐ Discrepancy ☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Peer Reach includes zero-Hit stations

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

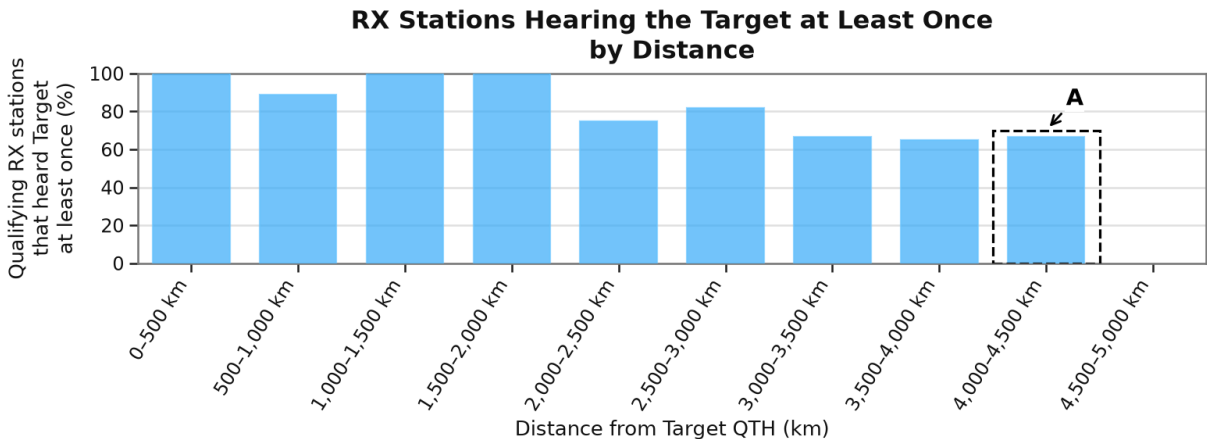
Peer Rx: AA9AL / FN41IM; KC2LFI / FN31EE; N1PQ / FN42FM; W1BW / FN42HL; W1KBX / FN43FE; WJ2V / FN30DO

Target TX KP4MD -> peer RX AA9AL, KC2LFI, N1PQ, W1BW, W1KBX, WJ2V

TX Performance | All directions; exact distance 4000-4500 km. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: Reach remains 4/6 = 66.6667%; pooled rate 30/277 = 10.8303% -> 34/281 = 12.0996%.



A marks the review area. Production plot elements; layout adapted for this card.

Peer RX / locator	Hits	Misses	Opportunities	Reached?
AA9AL / FN41IM	0	5	5	no
KC2LFI / FN31EE	2	80	82	yes
N1PQ / FN42FM	2	14	16	yes
W1BW / FN42HL	26	98	124	yes
W1KBX / FN43FE	0	7	7	no
WJ2V / FN30DO	4	43	47	yes

WHAT TO CHECK

1. All six receivers meet the five-opportunity threshold. Count four with one or more Hits.
2. Reach = 4/6. The separate pooled observation rate is 34/281; its denominator counts peer RX/cycle opportunities.
3. AA9AL and W1KBX contribute to reach and rates, but have no successful-SNR value; do not replace missing SNR with zero.

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Proposed answer: Peer Reach = $4/6 = 66.6667\%$; pooled Decode Rate = $34/281 = 12.0996\%$.

WSPRadar result: The native reach bar is 66.7%, based on 4 of 6 receivers. Both zero-Hit receivers remain eligible.

Protects: Distinguishes path reach from observation-level success and protects zero-Hit eligible peers.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P06 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

One complete three-hour bin

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

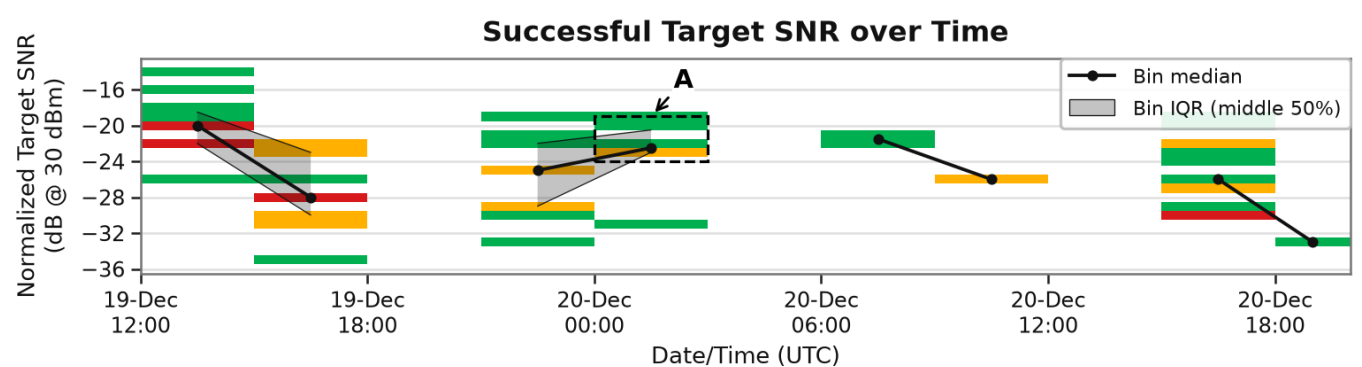
Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | VE6PDQ; 20 December 00:00-03:00 UTC. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 5/8 = 62.5% -> 6/9 = 66.6667%; median -22 -> -22.5 dB; the added 01:48 Target-only success has -31 dB.



A marks the review area. Production plot elements; layout adapted for this card.

20 Dec UTC	Outcome	SNR at 30 dBm	Target report ID	TX callsign	RX callsign
00:10	H	-23	44697734	KP4MD	VE6PDQ
00:12	M	-	-	no Target decode	VE6PDQ
00:22	H	-20	44698904	KP4MD	VE6PDQ
00:34	H	-22	44699745	KP4MD	VE6PDQ
00:56	H	-23	44701527	KP4MD	VE6PDQ
01:36	H	-19	44704914	KP4MD	VE6PDQ
01:48	H (Target-only)	-31	44705999	KP4MD	VE6PDQ
01:56	M	-	-	no Target decode	VE6PDQ
02:12	M	-	-	no Target decode	VE6PDQ

Report ID	UTC	TX callsign / locator	RX callsign / locator	Activity directly confirmed
44697961	20 Dec 00:12	N1PQ / FN42FM	VE6PDQ / DO34IR	Supporting TX N1PQ + peer RX VE6PDQ
44697864	20 Dec 00:12	KP4MD / CM98IQ	KZ3X / FN10WH	Target TX KP4MD + other RX KZ3X
44706605	20 Dec 01:56	WB2JEP / FM29LV	VE6PDQ / DO34IR	Supporting TX WB2JEP + peer RX VE6PDQ
44706574	20 Dec 01:56	KP4MD / CM98IQ	K6HVI / CN87VP	Target TX KP4MD + other RX K6HVI
44707778	20 Dec 02:12	WA5M / EM22JP	VE6PDQ / DO34IR	Supporting TX WA5M + peer RX VE6PDQ

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Report ID	UTC	TX callsign / locator	RX callsign / locator	Activity directly confirmed
44707791	20 Dec 02:12	KP4MD / CM98IQ	WB4KLJ / FM18KU	Target TX KP4MD + other RX WB4KLJ

WHAT TO CHECK

1. Count six Hits and three Misses: 6/9 = 66.6667%. One peer RX contributes a split vote of 2/3 Hit + 1/3 Miss; raw counts stay 6 and 3.
2. Include Target-only report 44705999 at 01:48. Sort all six SNRs: -31,-23,-23,-22,-20,-19. Median (-23-22)/2 = -22.5; Q1 -23; Q3 -20.5.

Proposed answer: Rate 66.6667%; opportunity count 9; station support 1.

Actual-SNR trace -22.5 dB with IQR [-23,-20.5] dB.

WSPRadar result: Production 00:00-03:00: 6 Hits, 3 Misses, 9 opportunities, 66.6667%; one station vote and the printed median/IQR.

Notes: All successful source rows: Target TX KP4MD/CM98IQ -> peer RX VE6PDQ/DO34IR. Miss rows have supporting other-TX receptions at VE6PDQ plus a separate Target TX activity witness; their full report IDs are retained in the independent native ledger.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P07 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason:

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

UTC folding at the end of the window

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | VE6PDQ; compare the 12 UTC and 20 UTC hour slots. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX. No-evidence date-hours are retained only in the folded display-date denominator; they are not invented opportunities.

Revision change: These two printed hour anchors remain numerically unchanged. Other hourly bins can change under opportunity-v3.

Date	Hour	Selected date-hour?	Hit	Miss	Peer RX present?
19 Dec	12 UTC	yes	0	0	0
20 Dec	12 UTC	yes	0	2	1
19 Dec	20 UTC	yes	0	2	1
20 Dec	20 UTC	outside window	0	0	0

WHAT TO CHECK

- 12 UTC occurs on both selected dates. Dec 19 has zero confirmed rows; it still belongs in the display denominator.
- For 12 UTC: (0+2) Misses / 2 dates = 1 mean Miss; (0+1) station presences / 2 = 0.5 mean station support.
- 20 UTC occurs only on Dec 19 because Dec 20 20:00 is the excluded end boundary. Divide its two Misses by one date, not two.

Proposed answer: 12 UTC: mean Misses 1, mean station support 0.5; 20 UTC: mean Misses 2, mean support 1.

Both rates are 0%; rates are calculated from pooled station outcomes, not averaged display counts.

WSPRadar result: Native folded profiles match both date denominators and mean support values in the table.

Protects: Detects both dropping zero-evidence dates and accidentally counting the end boundary as another selected hour.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P08 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Folded SNR: each date-hour gets one vote

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

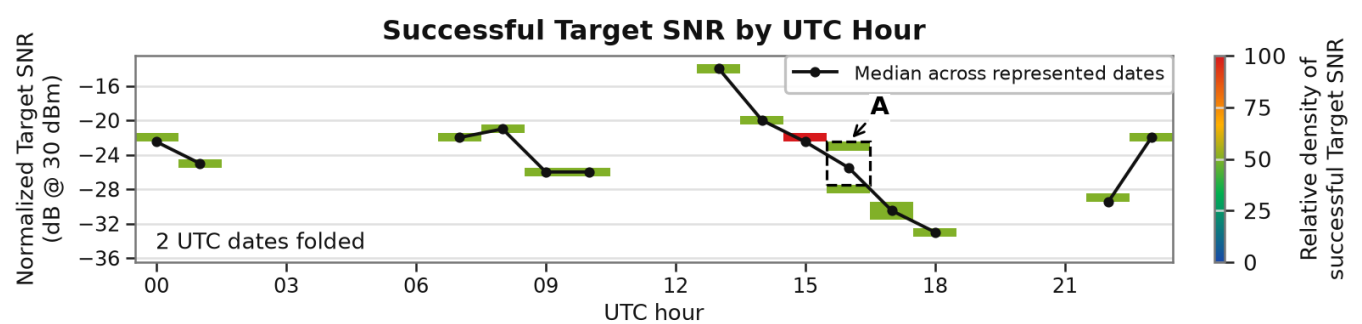
Peer Rx: VE6PDQ / DO34IR

Target TX KP4MD -> peer RX VE6PDQ / DO34IR

TX Performance | VE6PDQ; 16 UTC on 19 and 20 December. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 7 -> 9 successful samples; date medians [-28,-22] -> [-28,-23]; folded marker -25 -> -25.5 dB.



A marks the review area. Production plot elements; layout adapted for this card.

Date	Peer RX	Hits	Normalized successful SNRs (sorted)	Date-hour median
19 Dec	VE6PDQ / DO34IR	5	-30, -28, -28, -28, -26	-28
20 Dec	VE6PDQ / DO34IR	4	-27, -24, -22, -20	-23.0

Report ID	UTC	TX callsign	RX callsign	SNR at 30 dBm
44635714	19 Dec 16:12	KP4MD	VE6PDQ	-26
44636841	19 Dec 16:22	KP4MD	VE6PDQ	-28
44637729	19 Dec 16:30	KP4MD	VE6PDQ	-30
44638590	19 Dec 16:38	KP4MD	VE6PDQ	-28
44639716	19 Dec 16:48	KP4MD	VE6PDQ	-28
44770367	20 Dec 16:00	KP4MD	VE6PDQ	-20
44771374	20 Dec 16:12	KP4MD	VE6PDQ	-24
44773164	20 Dec 16:36	KP4MD	VE6PDQ	-22
44774245	20 Dec 16:50	KP4MD	VE6PDQ	-27

WHAT TO CHECK

1. First reduce each date-hour separately: 19 Dec median -28; 20 Dec median $(-24-22)/2 = -23$ dB.
2. Take the median of the two date-hour medians: $(-28-23)/2 = -25.5$ dB.
3. Pooling all nine raw successful SNR values instead gives -27 dB. The folded trace gives each represented date-hour one vote.

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Proposed answer: Folded marker -25.5 dB. Only two date-hour values contribute, below the five-value threshold for a visible temporal IQR band.

WSPRadar result: At 16 UTC, the native SNR marker is -25.5 dB; 2 date-hour values contribute, so no IQR band is visible.

Protects: A 1.5 dB contrast catches raw-observation pooling while checking the sparse-IQR display rule.

Notes: All listed reports directly confirm Target TX KP4MD/CM98IQ transmitted and peer RX VE6PDQ/DO34IR received. They include Target-only successes at 19 Dec 16:30 and 20 Dec 16:12.

Source: Complete 40 m archive capture; independently calculated ledger. All printed numerical anchors were compared with the WSPRadar pipeline.

YOUR DECISION - P09 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

SNR deviation: subtract each station baseline

Revised opportunity-v3 card: pending human review.

Analysis Direction: TX Performance

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

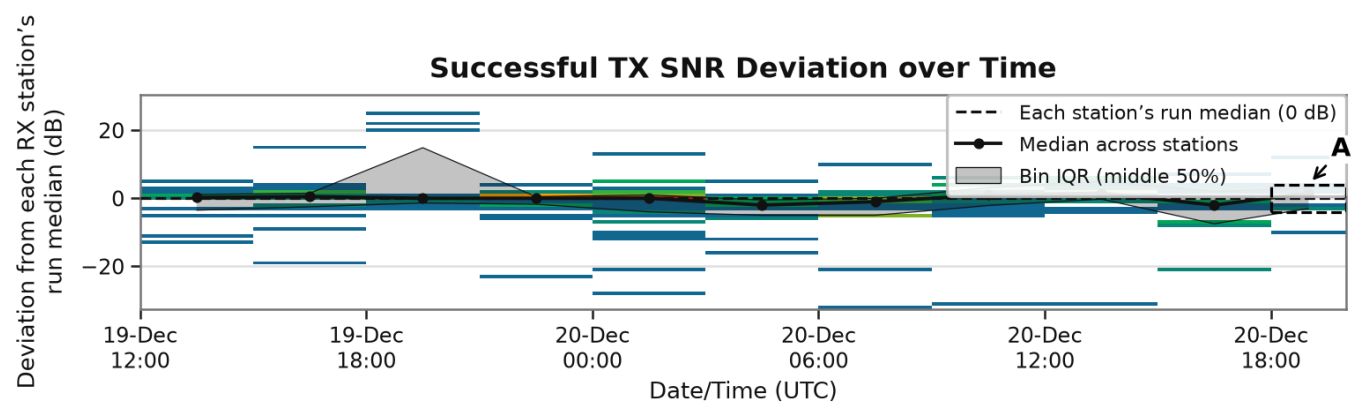
Peer Rx: VE6PDQ / DO34IR; N7NB / CN87UR; KE7MVX / CN83GG; VE7THZ / DN09FU; N7UL / CN84KA; NN6RF / CM87UW; N6GN / CM88OK; K6JEB / CM87UX; WB6RQN / CM98MQ

Target TX KP4MD -> peer RX VE6PDQ, N7NB, KE7MVX, VE7THZ, N7UL, NN6RF, N6GN, K6JEB, WB6RQN

TX Performance | Active scope; final partial bin, 20 December 18:00-20:00 UTC. Common setup on cover.

Activity evidence: A KP4MD decode confirms Target TX KP4MD and that exact peer RX. An unrelated TX decoded by that peer confirms that peer RX activity; a KP4MD report elsewhere confirms only Target TX activity, never a silent peer RX.

Revision change: 7 -> 9 contributors; median 0 -> +1.5 dB; IQR [-3.25,+1.75] -> [-3,+3] dB; VE6PDQ deviation -8 -> -10 dB.



A marks the review area. Production plot elements; layout adapted for this card.

Peer RX / locator	Baseline dB	Bin median dB	Deviation dB
VE6PDQ / DO34IR	-23	-33	-10
N7NB / CN87UR	-13.5	-17	-3.5
KE7MVX / CN83GG	-1	-4.0	-3.0
VE7THZ / DN09FU	-20	-22.0	-2.0
N7UL / CN84KA	-5	-3.5	1.5
NN6RF / CM87UW	-22.0	-20	2.0
N6GN / CM88OK	0	3.0	3.0
K6JEB / CM87UX	-5.0	-1	4.0
WB6RQN / CM98MQ	-29.0	-17	12.0

WHAT TO CHECK

- Each baseline is the full-window successful-SNR median, requiring at least three Hits. Include Target-only successes; subtract that peer RX baseline from its bin median.
- Each peer RX contributes once. Nine sorted deviations give median rank 5; linear Q1 is rank 3 and Q3 rank 7.

Target TX KP4MD / CM98 -> peer RX (identities on first card page)

Proposed answer: Active-scope anomaly median +1.5 dB, Q1 -3 dB, Q3 +3 dB; nine station values.

VE6PDQ contributes -33-(-23) = -10 dB. This differs from its actual-SNR trace at -33 dB.

WSPRadar result: The native final-bin marker is +1.5 dB, with IQR [-3,+3] dB and 9 contributing station values.

Notes: WB6RQN/CM98MQ is a peer RX on this TX Performance card: it received Target TX KP4MD. In the Benchmark cards WB6RQN has the separate Reference TX role; station identity alone does not determine report direction.

Source: Independent station baselines and bin summaries match production. All nine contributors are listed; end boundary 20:00 UTC remains excluded.

YOUR DECISION - P10 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Histogram weights and a positive outlier

Pending human review; no approval recorded.

Analysis Direction: TX Benchmark

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: 30 qualifying exact receiver identities; highlighted W6II / CN85NM

Reference Tx: WB6RQN / configured locator CM98 (raw reports CM98MQ)

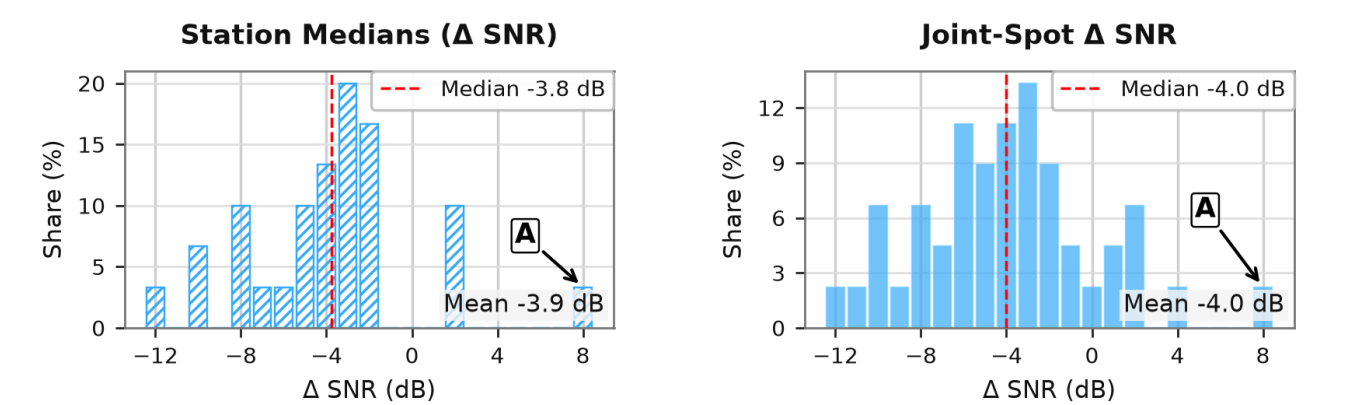
Target TX KP4MD -> peer RX W6II / CN85NM (highlighted +8 dB)

Reference TX WB6RQN -> peer RX W6II / CN85NM (same cycle)

Benchmark | Common setup on cover. All 30 qualifying receivers and 45 Joint observations.

Activity evidence: Each Target or Reference decode confirms its named TX transmitted and that exact peer RX received. Joint pairs require both TX reports at the same peer RX/full locator/cycle; unchanged Benchmark rules apply.

Revision change: Benchmark numerical anchors and methods unchanged; endpoint roles added.



Production plot elements; layout adapted for this card. A marks the +8 dB bars.

Distribution	Values counted	+8 dB entries	Central ordered value(s)
Station Medians	30	1	15th=-4; 16th=-3.5
Joint-Spot Delta SNR	45	1	23rd=-4

Report ID	UTC	TX callsign / locator	RX callsign / locator	Activity directly confirmed
44698886	20 Dec 00:22	KP4MD / CM98IQ	W6II / CN85NM	KP4MD transmitted; W6II received
44698891	20 Dec 00:22	WB6RQN / CM98MQ	W6II / CN85NM	WB6RQN transmitted; W6II received

WHAT TO CHECK

1. Locate the isolated +8 dB bar in each native histogram. Check its height using the two denominators below.
2. Read the two central ordered values for the station distribution and the central observation for the Joint distribution. Check the proposed medians.

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Proposed answer: Station +8 dB bar: $100/30 = 3.333\ldots\%$. Joint +8 dB bar: $100/45 = 2.222\ldots\%$.

Station median = $(-4 + -3.5)/2 = -3.75$ dB (native chart rounds to -3.8). Joint median = -4 dB.

WSPRadar result: Native histograms: one +8 dB entry in each; medians -3.8 dB (stations) and -4.0 dB (Joint).

Protects: Protects station weighting versus observation weighting, percent denominators, exact median calculation and preservation of the isolated positive extreme.

Notes: Human sign-off covers the selected bar and median arithmetic. Full 30/45 value lists and source IDs are independently calculated support, not claimed to have been individually reviewed.

+8 dB receiver: W6II (CN85nm), 2010-12-20T00:22:00+00:00. Source IDs 44698886 / 44698891.

Source: Outlier: W6II (CN85nm), 20 Dec 00:22 UTC; report IDs 44698886 / 44698891. Ordered-value lists retained in the candidate ledger.

YOUR DECISION - B02 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Decode Outcomes: which denominator?

Pending human review; no approval recorded.

Analysis Direction: TX Benchmark

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: 76 scoped receiver identities; example VE6PDQ / DO34IR

Reference Tx: WB6RQN / configured locator CM98 (raw reports CM98MQ)

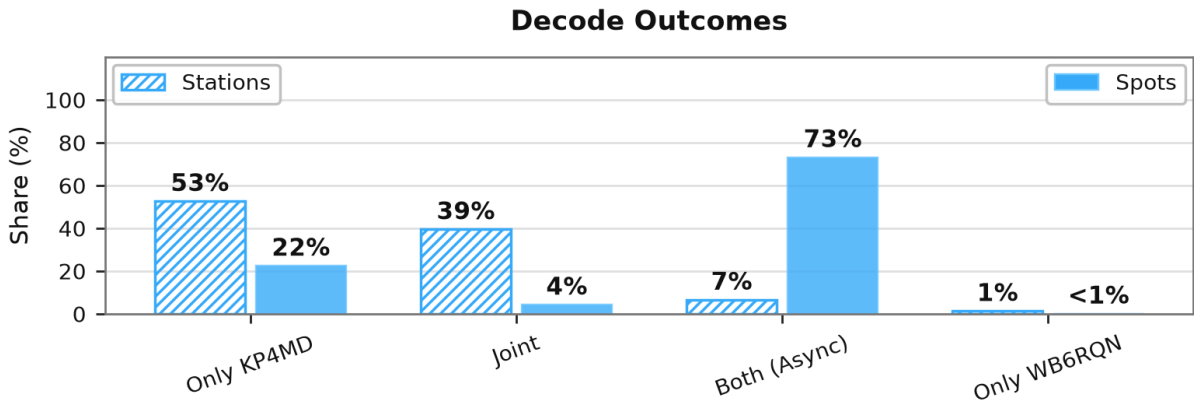
Target TX KP4MD -> scoped peer RX (example VE6PDQ / DO34IR)

Reference TX WB6RQN -> same scoped peer RX; Joint requires same cycle

Benchmark | Common setup on cover. Complete scoped evidence; include unpaired observations.

Activity evidence: Each Target or Reference decode confirms its named TX transmitted and that exact peer RX received. Joint pairs require both TX reports at the same peer RX/full locator/cycle; unchanged Benchmark rules apply.

Revision change: Benchmark numerical anchors and methods unchanged; endpoint roles added.



Production plot elements; layout adapted for this card.

Plotted category	Peer RX station count	Observation count
Only KP4MD	40	237
Joint	30	45
Both (Async)	5	771
Only WB6RQN	1	1
TOTAL	76	1054

WHAT TO CHECK

1. Check the Joint bars: divide 30 by 76 for stations and 45 by 1,054 for observations.
2. VE6PDQ has 56 non-joint observations and one Joint. Its station is Joint; its non-joint observations enter Both (Async). Confirm that distinction.

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Proposed answer: Joint stations: $100 \times 30/76 = 39.47\%$ (label 39%). Joint observations: $100 \times 45/1054 = 4.269\%$ (label 4%).

Do not insert raw 988 Only Target /45 Joint /21 Only Reference as the four plotted bars: these are cycle outcomes, while the visual categories also depend on station history.

WSPRadar result: Native labels: Joint stations 39%; Joint observations 4%; Both (Async) observations 73%.

Protects: Detects denominator swaps and confusing per-cycle outcomes with the native station-history composition.

Notes: These are composition percentages, not Decode Rates or Target success rates.

Supporting station examples (Only Target, Joint, Only Reference): VE6PDQ=56,1,0; K3TEZ=9,0,3; N7ZO=0,0,1.

Source: Frozen Milazzo raw reports; independent arithmetic and production replay. Full ledgers are retained with this review pack.

YOUR DECISION - B03 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Map value: two stations, three observations

Pending human review; no approval recorded.

Analysis Direction: TX Benchmark

Target Tx: KP4MD / configured locator CM98 (raw reports CM98IQ)

Peer Rx: VE6OG / DO33 and VE6PDQ / DO34IR

Reference Tx: WB6RQN / configured locator CM98 (raw reports CM98MQ)

Target TX KP4MD -> peer RX VE6OG / DO33 or VE6PDQ / DO34IR

Reference TX WB6RQN -> same peer RX and cycle

Benchmark | Common setup on cover. Map segment: 0-2,500 km, NNE (11.25 to 33.75 degrees).

Activity evidence: Each Target or Reference decode confirms its named TX transmitted and that exact peer RX received. Joint pairs require both TX reports at the same peer RX/full locator/cycle; unchanged Benchmark rules apply.

Revision change: Benchmark numerical anchors and methods unchanged; endpoint roles added.

Peer RX / locator	20 Dec UTC	Target TX KP4MD at 30dBm	Reference TX WB6RQN at 30dBm	Delta SNR
VE6OG DO33	00:22	-18.0	-22.0	4.0
VE6PDQ DO34ir	00:22	-20.0	-18.0	-2.0
VE6OG DO33	01:06	-22.0	-13.0	-9.0

Report ID	UTC	TX callsign / locator	RX callsign / locator	Activity directly confirmed
44698895	20 Dec 00:22	KP4MD / CM98IQ	VE6OG / DO33	KP4MD transmitted; VE6OG received
44698897	20 Dec 00:22	WB6RQN / CM98MQ	VE6OG / DO33	WB6RQN transmitted; VE6OG received
44698904	20 Dec 00:22	KP4MD / CM98IQ	VE6PDQ / DO34IR	KP4MD transmitted; VE6PDQ received
44698905	20 Dec 00:22	WB6RQN / CM98MQ	VE6PDQ / DO34IR	WB6RQN transmitted; VE6PDQ received
44702519	20 Dec 01:06	KP4MD / CM98IQ	VE6OG / DO33	KP4MD transmitted; VE6OG received
44702533	20 Dec 01:06	WB6RQN / CM98MQ	VE6OG / DO33	WB6RQN transmitted; VE6OG received

WHAT TO CHECK

1. Calculate the two station medians from the three listed Joint differences.
2. Take the median of those TWO station values. Compare with the production map value and contrast the pooled three-observation result.

Target TX KP4MD / CM98 -> peer RX (identities on first card page) | Reference TX WB6RQN / CM98

Proposed answer: VE6OG: median(-9,+4)=(-9+4)/2=-2.5 dB. VE6PDQ: one value=-2 dB.

Segment median=(-2.5 + -2)/2=-2.25 dB. WSPRadar stores SNR summaries to one decimal with ties-to-even rounding: -2.2 dB.

Pooled median(-9,-2,+4)=-2 dB, which is not the Benchmark map statistic. Support=2 qualifying station identities and 3 Joint observations.

WSPRadar result: Map: -2.2 dB from 2 qualifying stations and 3 Joint observations.

Protects: Checks the map uses equal station contributions and retains a distinction from the pooled Joint-Spot distribution.

Notes: Independent geography: VE6OG 1,772.78 km /17.455 degrees; VE6PDQ 1,895.44 km /15.320 degrees. Both lie within 0-2500 km and the NNE sector (11.25 to33.75 degrees).

Six original report IDs for the three pairs: 44698895/44698897; 44698904/44698905; 44702519/44702533.

Map value is an actual production segment-table output. The arithmetic table is a reviewer aid, not a native map rendering.

Source: VE6OG: 1,772.78 km / 17.455 degrees; VE6PDQ: 1,895.44 km / 15.320 degrees. Three pairs: IDs 44698895/44698897, 44698904/44698905, 44702519/44702533.

YOUR DECISION - B04 / revision 2

☐ Verified

☐ Discrepancy

☐ Unclear / skip

Correction / reason: _____
